International Journal of Sports Physiology and Performance, (Ahead of Print)
https://doi.org/10.1123/ijspp.2019-0395
© 2020 Human Kinetics, Inc.

Human Kinetics
ORIGINAL INVESTIGATION

No Influence of Prematch Subjective Wellness Ratings on External Load During Elite Australian Football Match Play

Phillip M. Bellinger, Cameron Ferguson, Tim Newans, and Clare L. Minahan

Purpose: To determine whether there is an association between external match load and contextual factors on subjective wellness in the days before and after Australian Rules football match play. ***Methods:*** A total of 34 elite male Australian football players completed a subjective wellness questionnaire in the days leading into the match (–3, –2, and –1 d), the day of (match day), and the days after each match (+3, +2, and +1 d). Players subjectively rated each item (mood, energy, stress, leg heaviness, muscle soreness, sleep quality, hours slept, and total wellness [a sum of the total response score]) on a visual analog scale ranging from 1 to 10, with 1 representing the negative end of the continuum. External load during competitive matches was quantified using accelerometer-derived PlayerLoad, and running activity was quantified using global positioning system technology across 2 competitive seasons. The relationships between perceptions of wellness (within-individual *z* score), external match load, and contextual factors (match result, match location, and between-matches recovery duration) were analyzed using linear mixed models. ***Results:*** Mixed-effect linear models revealed trivial effects of match-day wellness *z* score on subsequent external match load metrics. Match result (win) and PlayerLoad in the anteroposterior vector (au·min^{-1}) were associated with an increased (estimate ± SE: 0.30 ± 0.13 *z* score) and reduced subjective wellness (–0.15 ± 0.06 *z* score), respectively. ***Conclusion:*** The results of this study suggest that prematch perceived wellness does not relate to external match load in elite Australian football players. The between-matches microcycle length appears to be sufficient to restore perceived wellness to values that do not affect the subsequent external match loads.

Keywords: GPS technology, activity profile, training load, PlayerLoad

In team sports, the use of microtechnology, including global positioning systems (GPS) and triaxial accelerometers, is now important to monitor training and match movement patterns.[1,2] For example, information on athletes' activity profiles, such as total distance traveled and the magnitude of high-intensity running completed, quantified by GPS technology, and PlayerLoad, calculated from triaxial accelerometers, can be used to quantify external match and training loads.[1–3] An imbalance between training/competition loads and recovery over extended periods of time may contribute to reductions in training output[4] and possibly result in overtraining symptoms.[5] This has resulted in an increase in attention being given to the evaluation of monitoring tools which may indicate the fatigue status of athletes.[5]

A number of monitoring tests exist that assess the presence and magnitude of athlete fatigue, with those selected needing to be valid, reliable, and practically convenient in an applied setting.[5,6] Although many customized "wellness" questionnaires have not undergone a rigorous process of validation, because of their ease of administration and sensitivity to changes in performance,[7–9] practitioners have incorporated wellness questionnaires into their monitoring practices.[4,10–12] As such, it is thought that subjective wellness scales represent a time-efficient and noninvasive method for practitioners to gain information related to a player's wellness status and his/her overall readiness to train and compete.[11] There is a growing body of evidence suggesting that a reduction in subjective wellness has a negative impact on the subsequent external load of training.[4,10] Recently, Gallo et al[4] were able to demonstrate that a reduced wellness *z* score corresponded to the impairments in accelerometer-derived estimates of external load during training in Australian football players. Furthermore, it has also been reported[10] that reductions in perceptual wellness reduce numerous metrics of external training load in elite male soccer players. However, it remains to be determined whether team-sport athletes similarly regulate their running and movement intensity during match play as they do in training in response to changes in subjective wellness.

Only one previous study[13] has examined the associations between the changes in prematch wellness and changes in match running performance during competitive competition. Ihsan et al[13] reported that when normalized to playing time, changes in prematch wellness had a significant *very large* correlation, with changes in total distance covered over the course of a field hockey tournament that consisted of 6 matches played over 9 days. Furthermore, when normalized to rating of perceived exertion (RPE), the change in prematch wellness demonstrated very large to almost perfect (–0.89 to –0.95) associations, with changes in high-intensity acceleration, deceleration, and running distance covered. Although these findings are important, the congested match fixture is not representative of an Australian Football League (AFL) competitive season fixture (typically 6–8 d between-match recovery periods), and the GPS-derived running-based measures[13] may not reflect the low-velocity movements, changes of direction, and physical contact that occur frequently in contact sports, such as Australian rules football.[14] Indeed, accelerometer-derived metrics of PlayerLoad may provide a more precise representation of these movements compared with GPS-derived metrics. In particular, PlayerLoad$_{SLOW}$, which is the PlayerLoad accumulated during activities when running speed is <2 m·s^{-1}, has been shown to accumulate to a greater extent

Bellinger, Newans, and Minahan are with Griffith Sports Physiology and Performance, School of Allied Health Sciences, and Bellinger also Menzies Health Inst Queensland, Griffith University, Gold Coast, QLD, Australia. Ferguson is with the High Performance Unit, Manly Warringah Sea Eagles, Sydney, NSW, Australia. Bellinger (p.bellinger@griffith.edu.au) is corresponding author.

Authenticated t_lambrinides@yahoo.com | Downloaded 03/18/20 12:36 PM UTC

during contact, compared with noncontact small-sided games.[15] This suggests that $PlayerLoad_{SLOW}$ is capable of quantifying the collision aspect of match play. As such, it remains to be determined whether these important physical parameters of Australian rules match play are affected by prematch perceptions of wellness across an entire season, as opposed to a multiday tournament,[13] and whether contextual factors (match result, match location, and between-match recovery duration) may mediate this relationship. Therefore, the purpose of the present study was to determine whether there is an association between prematch and postmatch self-reported perceptions of wellness and the influence of contextual factors on the external load of elite Australian football players during match play across 2 competitive seasons.

Methodology

Participants

In total, 34 elite male Australian football players (mean [SD]: stature 188.2 [7.3] cm, body mass 87.0 [8.1] kg, age 22.1 [3.3] y) from the same team gave written consent to participate in this study, which was conducted across 2 full seasons of the AFL competition. Based on previous research,[16] inclusion criteria were that players selected for analysis had previously played a minimum of 20 AFL matches and played a minimum of 75% match time within the given match of interest. A total of 460 matches and 1840 self-reported perceptions of wellness were included in the analysis. Ethical approval to conduct this study was obtained from the Griffith University's Human Ethics Committee.

Study Design

The present study was a retrospective observational design where we aimed to determine whether there is an association between prematch and postmatch wellness, external match load metrics, and contextual factors (match result, match location, and between-match recovery duration) with linear mixed models.

Subjective Wellness Questionnaire

Players were instructed to complete a customized subjective wellness questionnaire on the days leading into the match (−3, −2, and −1 d), the day of (match day), and the days following (+3, +2, and +1 d) each respective match irrespective of microcycle length (ie, between match recovery duration). There were four 6-day, thirty 7-day, six 8-day, one 14-day, and one 15-day between-match recovery periods. The questionnaire was designed to be short and specific and was based on the common items featured on wellness questionnaires that are reported in the literature.[4,7,8,11] The questionnaire assessed the following elements of wellness: (1) mood, (2) energy, (3) stress, (4) leg heaviness, (5) muscle soreness, (6) sleep quality, (7) hours slept, and (8) total wellness (a sum of the total response score). Each item was assessed on a visual analog scale ranging from 1 to 10, with 1 representing the negative end of the continuum. The subjective wellness questionnaire was completed on each player's mobile phone and collated in a central database using customized athlete data management software (Smartabase, Fusion Sport, Brisbane, Australia). The central database was checked daily to ensure compliance was met. For each day, players were required to complete their wellness entries before the team meeting (conducting at 8:00 AM) on training days or before 12:00 PM on days off (3 d before each match). Within-player z scores were calculated using the following formula: (individual player's score − individual player's average)/individual player's SD.

External Match Load

During each match, players wore a commercially available GPS technology device (MinimaxX, Team 2.5; Catapult Innovations, Scoresby, Australia) sampled at a rate of 10 Hz. Each device also housed a 100-Hz accelerometer, determining various PlayerLoad metrics (Catapult Innovations, Melbourne, Australia) that have been shown to have high levels of relevance and reliability in team-sport-specific movements.[3,14,17] When selected, each player wore the same unit for all the matches. The device was placed in a customized vest sitting between the shoulder blades of the participants, restricting movement artifact. Following each match, the data were downloaded using Logan Plus software (version 4.0; Catapult Innovations, Melbourne, Australia) and were to include time spent on the field of play based on substitution times manually recorded during the game. The data from each of the 3 vectors (x, y, and z axes) were quantified independently as $PlayerLoad_{AP}$ (ie, anteroposterior), $PlayerLoad_{ML}$ (mediolateral), and $PlayerLoad_{V}$ (vertical). The accumulated data from all the 3 axes were integrated to formulate the vector magnitude, which the manufacturers of the MinimaxX accelerometers have titled as "PlayerLoad." The accumulated data from the mediolateral and anterioposterior axes were equated to quantify $PlayerLoad_{2D}$ (2-dimensional), whereas data from all vectors when movement was <2 $m \cdot s^{-1}$ were used to describe low-velocity external load ($PlayerLoad_{SLOW}$). Activity profile was also quantified by assessing total distance covered (in meters) and total high-speed running distance (≥24.0 km h^{-1}). All external match load metrics were divided by the time spent on the ground for each individual player. As such, external match load was measured using PlayerLoad (arbitrary units [au]·min^{-1}), $PlayerLoad_{SLOW}$ (au·min^{-1}), $PlayerLoad_{2D}$ (au·min^{-1}), $PlayerLoad_{AP}$ (au·min^{-1}), $PlayerLoad_{ML}$ (au·min^{-1}), $PlayerLoad_{V}$ (au·min^{-1}), relative distance (m·min^{-1}), and high-speed distance (m·min^{-1}; ≥24.0 km·h^{-1}).

Statistical Analysis

To examine the effect of wellness z score on the external load parameters, linear mixed models were performed using the lme4[18] and afex packages in RStudio statistical software (version 0.99.903; RStudio Inc, Boston, MA). Linear mixed models were deemed as the appropriate statistical method to employ, given that our clustered data set arose from repeated samples from the same subject across numerous time points.[19] To model the external match load within each match, the location of each match (ie, home or away) and the between-match microcycle length (ie, days of break between matches) were treated as fixed effects, whereas the player and match were treated as random effects. When modeling the 1-day postmatch wellness z score, the same external match load metrics were used, with the result (ie, win or loss) inserted as a fixed effect as well. When modeling the 1-day postwellness mixed model, models were compared using the Bayesian information criterion, with the lowest Bayesian information criterion being deemed parsimonious. Differences in wellness z scores between different microcycle durations were also evaluated using the effect size statistic with upper and lower 95% confidence intervals, which were classified as trivial (−0.2 to 0.2), small (0.2 to 0.6), moderate (0.6 to 1.2), large (1.2 to 2.0), and very large (>2.0).

Authenticated t_lambrinides@yahoo.com | Downloaded 03/18/20 12:36 PM UTC

Results

The mean relative external load (SD) during each match was 12.6 (1.24) au·min^{-1} PlayerLoad, 4.24 (0.90) au·min^{-1} PlayerLoad$_{SLOW}$, 8.36 (0.89) au·min^{-1} PlayerLoad$_{2D}$, 5.82 (0.71) au·min^{-1} PlayerLoad$_{AP}$, 4.71 (0.59) au·min^{-1} PlayerLoad$_{ML}$, 8.12 (0.91) au·min^{-1} PlayerLoad$_{V}$, 125.1 (12.6) m·min^{-1} relative distance, and 0.53 (0.47) m·min^{-1} total high-speed distance.

Figure 1 presents the wellness z scores for 6-, 7-, and 8-day microcycles in the days leading up to the match (−3, −2, and −1 d), the day of (match day), and the days following each match (+3, +2, and +1 d). There were very large differences between match day (ie, 0 d) wellness z score and day 1 postmatch for each microcycle duration ($d = 5.0$–6.6) (Table 1). The day following each match (ie, +1 d) represented the lowest wellness z score for each microcycle duration. Perceptions of wellness showed small improvements in the days leading up to each match (−2 and −1) and showed large and very large improvements on the day of each match (match day) and in the days after each match (+2 and +3), respectively. During a 6-day microcycle, the wellness z score of −3 days prematch (z score ± 90% confidence limit: 0.01 ±0.16 au) was moderately lower compared with both a 7- (0.25 ± 0.07 au; $d = 0.71$) and 8-day (0.27 ± 0.09 au; $d = 0.75$) microcycle, whereas during an 8-day microcycle, the wellness z score of +1 day postmatch (−0.98 ± 0.22 au) was higher compared with a 7-day microcycle (−1.45 ± 0.11 au; $d = 1.37$). In addition, an 8-day microcycle had a higher wellness z score for +2 days postmatch (−0.22 ± 0.18 au) compared with a 6-day microcycle (−0.63 ± 0.13 au; $d = 0.95$) and a 7-day microcycle (−0.54 ± 0.09 au; $d = 1.23$).

Linear mixed model parameter estimates assessing the relationship between match-day subjective wellness and external match load are presented in Table 2. The model revealed that prematch wellness z score, as well as all individual wellness items, had trivial effects on all external match load metrics. In converse, match result (win) and PlayerLoad$_{AP}$ had a substantial impact on perceptions of wellness on the day following each match (Table 3). A win resulted in a 0.30 ± 0.13 (estimate ± SE) z score increase in subjective wellness, whereas every 1-unit increase in PlayerLoad$_{AP}$ per minute was associated with 0.15 ± 0.06 z score reduction in subjective wellness. Furthermore, an increase in high-speed distance (m·min^{-1}; ≥24.0 km·h^{-1}) was associated with a 0.26 ± 0.10 increase in subjective wellness.

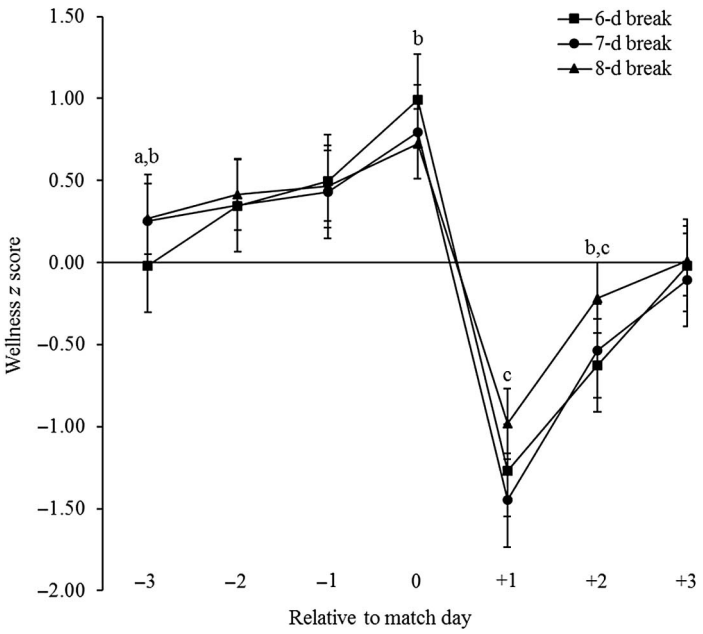

Figure 1 — Self-reported subjective wellness z score on the days leading into the match (−3, −2, and −1 d), the day of (match day), and the days following each match (+3, +2, and +1 d) for a 6-, 7-, and 8-day microcycle. [a] Significantly different 7- versus 6-day microcycle. [b] Significantly different 8- versus 6-day microcycle. [c] Significantly different 8- versus 7-day microcycle.

Discussion

The main purpose of the present study was to establish whether elite Australian football players regulate their running and movement intensity during match play in response to alterations in prematch subjective wellness and whether contextual factors (match result, match location, and between-match recovery duration) mediate this association. There were trivial associations between prematch wellness z score, as well as all individual wellness items, on external match load metrics. However, the match result (win) and a higher PlayerLoad$_{AP}$ were associated with an increase and reduced subjective wellness rating on the day following each match, respectively. The restored wellness values on match day suggest that the between-match recovery duration (typically 6–8 d) is sufficient to eliminate the effect of fatigue on external match load.

In the present study, prematch subjective wellness was not significantly related to any of the GPS- or accelerometer-derived metrics of exercise intensity during AFL match play. These findings contrast with other literature that has shown that reductions in pretraining subjective wellness reduce the subsequent external load during training in elite team-sport athletes.[4,10,12] In elite Australian football players, Gallo et al[4] demonstrated that a wellness z score of −1 corresponded to a significant reduction of −4.9% ± 3.1% and −8.6% ± 3.9% in PlayerLoad and PlayerLoad$_{SLOW}$, respectively, compared with those without reduced wellness. In elite soccer players, Malone et al[10] reported that a pretraining wellness z score of −1 had a significant negative impact on total high-speed distance, maximal velocity, maximal velocity exposures, PlayerLoad, and PlayerLoad$_{slow}$, compared with those without reduced wellness. Finally, Govus et al[12] concluded that a higher wellness z score (ie, the player felt better overall) was associated with a higher PlayerLoad during training in American Football players. Although these studies[4,10,12] concluded that reductions in pretraining subjective wellness may directly affect the player's ability to reach maximal running velocities, accumulate distance at high velocities, and accumulate accelerometer-derived PlayerLoad metrics, we did not find a similar relationship between prematch subjective wellness and subsequent external load during match play. There are likely 2 reasons that could support this finding: (1) the between-match recovery duration (typically 6–8 d) is sufficient to restore subjective wellness to values that do not impact upon the subsequent external match loads or (2) players do not regulate their running and movement intensity as they do in training due to the heightened motivation of competitive match play. The results from the present study, and that of others,[7,8,20] show that subjective wellness is typically impaired for 72 to 96 hours postmatch, and peak subjective wellness values are rated on the day of each match. As such, there are only small variations in subjective wellness on match day, and this small variation does not seem to be related to external match load. In contrast, given that

Authenticated t_lambrinides@yahoo.com | Downloaded 03/18/20 12:36 PM UTC

Table 1 Self-Reported Subjective Well-Being *z* Score on the Days Leading Into the Match (−3, −2, and −1 d), the Day of (Match Day), and the Days Following Each Match (+3, +2, and +1 d) for a 6-, 7-, and 8-Day Microcycle

Relative to match day, d	6-d microcycle	7-d microcycle	Δ7- to 6-d microcycle	8-d microcycle	Δ8- to 6-d microcycle	Δ8- to 7-d microcycle
−3	0.01 (−0.14 to 0.17)	0.25 (0.19 to 0.32)[a]	0.87 (0.27 to 1.43); moderate increase	0.27 (0.17 to 0.36)[b]	0.77 (0.25 to 1.27); moderate increase	Unclear
−2	0.35 (0.21 to 0.49)	0.35 (0.27 to 0.43)	Unclear	0.41 (0.32 to 0.51)	Unclear	Unclear
−1	0.50 (0.37 to 0.62)	0.43 (0.37 to 0.49)	Unclear	0.47 (0.37 to 0.56)	Unclear	Unclear
Match day	0.99 (0.81 to 1.18)	0.80 (0.67 to 0.92)	Unclear	0.72 (0.57 to 0.88)[b]	−0.53 (−0.93 to −0.07); small decrease	Unclear
+1	−1.27 (−1.59 to −0.94)	−1.45 (−1.54 to −1.35)	Unclear	−0.98 (−1.20 to −0.76)[c]	Unclear	1.11 (0.64 to 1.53); moderate increase
+2	−0.63 (−0.76 to −0.50)	−0.54 (−0.62 to −0.45)	Unclear	−0.22 (−0.40 to −0.04)[b,c]	0.98 (0.46 to 1.47); moderate increase	0.90 (0.40 to 1.36); moderate increase
+3	0.02 (−0.14 to 0.11)	−0.11 (−0.17 to −0.05)	Unclear	−0.01 (−0.10 to −0.12)	Unclear	Unclear

Note: Data are presented as effect-size difference (95% confidence interval) and qualitative classification.
[a] Significantly different 7- versus 6-day microcycle. [b] Significantly different 8- versus 6-day microcycle. [c] Significantly different 8- versus 7-day microcycle.

Authenticated t_lambrinides@yahoo.com | Downloaded 03/18/20 12:36 PM UTC

Table 2 Linear Mixed-Model Parameter Estimates for the Effect of Match-Day Subjective Wellness on External Match Load

Metric	Intercept ± SE	Coefficient ± SE	*P*
Relative distance, m·min^{-1}	132.31 ± 2.02	1.14 ± 0.96	.238
PlayerLoad, au·min^{-1}	12.48 ± 0.20	0.12 ± 0.09	.183
PlayerLoad$_{2D}$, au·min^{-1}	8.29 ± 0.15	0.05 ± 0.06	.449
PlayerLoad$_{SLOW}$, au·min^{-1}	3.75 ± 0.15	–0.00 ± 0.09	.997
PlayerLoad$_{AP}$, au·min^{-1}	5.73 ± 0.14	0.07 ± 0.05	.205
PlayerLoad$_{ML}$, au·min^{-1}	4.63 ± 0.10	0.07 ± 0.04	.102
PlayerLoad$_{V}$, au·min^{-1}	7.97 ± 0.17	0.10 ± 0.06	.130
High-speed distance (≥24.0 km·h^{-1}), m·min^{-1}	0.44 ± 0.07	0.07 ± 0.04	.108

Abbreviations: AP, anteroposterior; ML, mediolateral; SLOW, low-velocity external load; V, vertical.

Table 3 Linear Mixed-Model Parameter Estimates for the Effect of External Match Load on Subjective Wellness *z* Score 1 Day Postmatch

Metric	Reduced model	Full model
Intercept	–1.17 ± 0.38**	–1.40 ± 0.65
Relative distance, m·min^{-1}	—	0.00 ± 0.01
PlayerLoad, au·min^{-1}	—	0.29 ± 0.97
PlayerLoad$_{2D}$, au·min^{-1}	—	–0.55 ± 1.65
PlayerLoad$_{SLOW}$, au·min^{-1}	—	0.02 ± 0.06
PlayerLoad$_{AP}$, au·min^{-1}	–0.15 ± 0.06*	0.11 ± 1.11*
PlayerLoad$_{ML}$, au·min^{-1}	—	0.24 ± 0.99
PlayerLoad$_{V}$, au·min^{-1}	—	–0.23 ± 0.75
High-speed distance (≥24.0 km·h^{-1}), m·min^{-1}	0.26 ± 0.10*	0.21 ± 0.11*
Result (win)	0.30 ± 0.13*	0.34 ± 0.13*
Break	—	0.00 ± 0.12
Location (home)	—	–0.23 ± 0.13
Bayesian information criterion	834.8	879.6

Abbreviations: AP, anteroposterior; ML, mediolateral; SLOW, low-velocity external load; V, vertical. Note: Data are presented as estimate ± SE.
*$P < .05$. **$P < .01$.

there are large impairments in subjective wellness in the days following each match[7,8,20] and during the precompetition phase of the season,[4] the large variation in pretraining subjective wellness during these periods appears to have a larger impact on the subsequent external output during training.

Although the majority of previous research has focused on investigating the impact of impaired pretraining perceptual wellness ratings on the subsequent external output during training,[4,10,12] only one study[13] has investigated whether alterations in prematch perceptual wellness modify exercise intensity during match play. In contrast to the findings of the present study, Ihsan et al[13] reported that changes in wellness were associated with alterations in GPS-derived metrics of match running performance throughout a 6-match field hockey tournament. Interestingly, Ihsan et al[13] reported that changes in prematch wellness had a significant very large correlation, with changes in total distance covered when normalized to playing time. However, when normalized to RPE, the change in prematch wellness demonstrated very large to almost perfect (–0.89 to –0.95) associations, with changes in high-intensity acceleration, deceleration, and running distance covered. As such, the relationship between changes in prematch perceptual wellness and the subsequent external output of particular variables when normalized to either time or RPE is still not clear. Furthermore, Ihsan et al[13] studied male field hockey players throughout 6 matches played over 9 days, whereby the changes in wellness and match load were highly variable between matches (small to large effect size changes). Given the major differences in the tournament schedule[13] compared with the AFL competitive season, whereby players typically have a between-match recovery duration of 6 to 8 days, it is likely that the longer between-match recovery periods during an AFL competitive season were sufficient to restore subjective wellness to values that do not impair the external load during matches. Furthermore, the external load metrics were resigned to GPS-derived measures of running intensity,[13] which is not indicative of Australian rules football as it involves frequent low-velocity movements, changes of direction, and physical contact, which common GPS-derived running metrics do not accurately represent.[14,17] However, despite the importance of overall PlayerLoad and the way in which PlayerLoad is accumulated,[14] we did not find a significant relationship between prematch subjective wellness and the subsequent accelerometer-derived metrics of external match load. In contrast to these findings, Mooney et al[3] established that AFL players who had a substantially impaired neuromuscular status (assessed by flight time to contraction

Authenticated t_lambrinides@yahoo.com | Downloaded 03/18/20 12:36 PM UTC

time ratio obtained from a countermovement jump)[21] accumulated PlayerLoad with altered mechanics compared with nonfatigued players, and this change was perceived negatively by coaches. Furthermore, Rowell et al[22] recently established that elite Australian soccer players with a reduced flight time to contraction time ratio had increased $PlayerLoad_{ML}$ (in percentage) and decreased $PlayerLoad_{V}$ during small-sided game play, further demonstrating fatigue-driven modification to movement strategy in team-sport athletes. However, given that there were large reductions in the flight time to contraction time in both of these studies,[3,22] compared with the peak wellness values evident on match day in the current study, it is likely that this diluted any evidence of an impairment in subjective wellness having a negative effect on the subsequent external match load in the context of the present study.

In the present study, there was an observed daily fluctuation in perceived ratings of physical and psychological wellness in the days before and after matches, with the greatest and least changes reported on the day following a match and the days prior to a match, respectively. The magnitude of reduction in postmatch subjective wellness did not seem to be related to microcycle duration (ie, between-match recovery duration) or match location (ie, home or away match). Furthermore, there were unclear differences in perceptual wellness in the 2 days before each match, independent of microcycle duration (ie, number of between-match recovery days). These findings are supported by previous research that has assessed various recovery indices following Australian rules football match play, demonstrating that measures of perceptual,[8,20] neuromuscular,[23] biochemical,[23] and immune markers[24] return to baseline within 72 to 96 hours postmatch in elite Australian football players.

A novel finding from the present study was that the match result (ie, win or loss) had a substantial impact on perceptions of wellness in the day following each match, with a win resulting in a 0.30 ± 0.13 z score increase in subjective wellness. Increased ratings of mood and energy were responsible for the higher subjective wellness in the day following a successful outcome, coupled with lower ratings of sleep quality and quantity following a loss. Conceivably, perceived stress and anxiety that may have been associated with an unsuccessful match outcome,[25,26] coupled with the higher levels testosterone concentration following a win relative to a loss,[27,28] may have been related to the alterations in subjective wellness following differential match outcomes in the current study. Indeed, a classical report[26] detailing the emotional responses from rugby players demonstrated a range of pleasant emotional states following a win and unpleasant emotions after a loss. Furthermore, alterations in testosterone levels were positive following successful match outcomes and negative following unsuccessful match outcomes in female soccer players.[27] In the present study, the substantial impact of match result on perceptions of wellness on the day following each match may be related to the affective emotional responses driven by match outcome or the subsequent physiological changes following a differential match outcome.

The external match load variables that related to subjective wellness on the day following each match were $PlayerLoad_{AP}$ and high-speed running distance. Greater accumulation of $PlayerLoad_{AP}$ (ie, 1-unit increase) during matches resulted in a reduction in subjective wellness (–0.15 z score) on the day following each match. Given that $PlayerLoad_{AP}$ provides a summation of the rate of acceleration change that occurred during activity within the anteroposterior plane, it may be that greater volumes of work completed within this type of action are related to postmatch fatigue and/or recovery and, as a result, subjective wellness. Indeed, one previous study[29] showed that there were moderate to large positive correlations between the distance traveled by accelerating (3 to 15 $m \cdot s^{-2}$) and decelerating (–3 to –15 $m \cdot s^{-2}$) and the increase in creatine kinase 24 hours postmatch. As such, it may be that greater accumulation of $PlayerLoad_{AP}$ may lead to a greater magnitude of muscle damage, which could influence subjective wellness. Another interesting finding of the present study was that an increase in high-speed distance resulted in a 0.30 ± 0.13 z score increase in subjective wellness. This would seem counterintuitive, given that greater accumulation in $PlayerLoad_{AP}$ was associated with a reduction in wellness. Although speculative, one hypothesis that may explain this finding is that the increase in high-speed distance in particular matches may have been indicative of a less contested/physical match, and the reduction in physicality may have alleviated the reduction in subjective wellness on the day following each match. Unfortunately, we do not have access to the direct quantification of the collisions from the current data set and cannot confirm this hypothesis directly. Nonetheless, these findings provide further evidence to suggest that the external load from match play may influence the subjective wellness responses on the subsequent day.

Although the present study was conducted in an elite sporting environment, it also has limitations. The design of the subjective wellness questionnaire employed in this study was based on previous literature,[4,7,8,11] but it had not undergone a rigorous process of development and evaluation to ensure that the responses would have an acceptable degree of validity and reliability. Furthermore, training load prescription in the present study was not controlled by the research team.

Conclusion

The results of this study suggest that prematch perceived wellness does not substantially impact the subsequent external match load in elite Australian football players. Interestingly, the majority of external match load metrics did not seem to be related to perceptions of wellness on the day following each match either. However, match result (win) and $PlayerLoad_{AP}$ had a substantial impact on perceptions of wellness on the day following each match with a win resulting in a 0.30 ± 0.13 z score increase in subjective wellness, whereas every 1-unit increase in $PlayerLoad_{AP}$ per minute was associated with a 0.15 ± 0.06 z score reduction in subjective wellness.

Practical Application

Although there did not seem to be a strong relationship between prematch perceived wellness and the subsequent external output from elite Australian football players during match play in the current study, it is likely that the between-match recovery duration (typically 6–8 d) is sufficient to restore subjective wellness. As such, we would still promote the use of subjective wellness assessments during the season, as this still may allow for the identification of a player who is "at risk" of a reduced training or match exercise intensity, and appropriate recovery strategies or reduced training leading up to the match could be incorporated in an attempt to increase perceptual wellness for that player. It is important to consider wellness responses in the context of each player, and we suggest using relative changes in wellness

Authenticated t_lambrinides@yahoo.com | Downloaded 03/18/20 12:36 PM UTC

(ie, wellness z scores) to interpret the individual responses to subjective wellness.

References

1. Aughey RJ. Applications of GPS technologies to field sports. *Int J Sports Physiol Perform*. 2011;6(3):295–310. doi:10.1123/ijspp.6.3.295
2. Cummins C, Orr R, O'Connor H, West C. Global positioning systems (GPS) and microtechnology sensors in team sports: a systematic review. *Sports Med*. 2013;43(10):1025–1042. PubMed ID: 23812857 doi:10.1007/s40279-013-0069-2
3. Mooney MG, Cormack S, O'Brien BJ, Morgan WM, McGuigan M. Impact of neuromuscular fatigue on match exercise intensity and performance in elite Australian football. *J Strength Cond Res*. 2013;27(1):166–173. PubMed ID: 22395264 doi:10.1519/JSC.0b013e3182514683
4. Gallo TF, Cormack SJ, Gabbett TJ, Lorenzen CH. Pre-training perceived wellness impacts training output in Australian football players. *J Sports Sci*. 2016;34(15):1445–1451. PubMed ID: 26637525 doi:10.1080/02640414.2015.1119295
5. Halson SL. Monitoring training load to understand fatigue in athletes. *Sports Med*. 2014;44(suppl 2):139–147. doi:10.1007/s40279-014-0253-z
6. Twist C, Highton J. Monitoring fatigue and recovery in rugby league players. *Int J Sports Physiol Perform*. 2013;8(5):467–474. doi:10.1123/ijspp.8.5.467
7. Gastin PB, Meyer D, Robinson D. Perceptions of wellness to monitor adaptive responses to training and competition in elite Australian football. *J Strength Cond Res*. 2013;27(9):2518–2526. PubMed ID: 23249820 doi:10.1519/JSC.0b013e31827fd600
8. Gallo TF, Cormack SJ, Gabbett TJ, Lorenzen CH. Self-reported wellness profiles of professional Australian football players during the competition phase of the season. *J Strength Cond Res*. 2017;31(2):495–502. PubMed ID: 27243912
9. Gallo T, Cormack S, Gabbett T, Williams M, Lorenzen C. Characteristics impacting on session rating of perceived exertion training load in Australian footballers. *J Sports Sci*. 2015;33(5):467–475. PubMed ID: 25113820 doi:10.1080/02640414.2014.947311
10. Malone S, Owen A, Newton M, et al. Wellbeing perception and the impact on external training output among elite soccer players. *J Sci Med Sport*. 2017;21(1):29–34. PubMed ID: 28442275 doi:10.1016/j.jsams.2017.03.019
11. Taylor K, Chapman D, Cronin J, Newton MJ, Gill N. Fatigue monitoring in high performance sport: a survey of current trends. *J Aust Strength Cond*. 2012;20(1):12–23.
12. Govus AD, Coutts A, Duffield R, Murray A, Fullagar H. Relationship between pretraining subjective wellness measures, player load, and rating-of-perceived-exertion training load in American college football. *Int J Sports Physiol Perform*. 2018;13(1):95–101. doi:10.1123/ijspp.2016-0714
13. Ihsan M, Tan F, Sahrom S, Choo HC, Chia M, Aziz AR. Pre-game perceived wellness highly associates with match running performances during an international field hockey tournament. *Eur J Sport Sci*. 2017;17(5):593–602. PubMed ID: 28398168 doi:10.1080/17461391.2017.1301559
14. Boyd LJ, Ball K, Aughey RJ. Quantifying external load in Australian football matches and training using accelerometers. *Int J Sports Physiol Perform*. 2013;8(1):44–51. doi:10.1123/ijspp.8.1.44
15. Boyd L, Gallaher E, Ball K, Stepto N, Aughey R, Varley M. Practical application of accelerometers in Australian football. *J Sci Med Sport*. 2010;13:e14–e15. doi:10.1016/j.jsams.2010.10.491
16. Coutts AJ, Quinn J, Hocking J, Castagna C, Rampinini E. Match running performance in elite Australian rules football. *J Sci Med Sport*. 2010;13(5):543–548. PubMed ID: 19853508 doi:10.1016/j.jsams.2009.09.004
17. Boyd LJ, Ball K, Aughey RJ. The reliability of MinimaxX accelerometers for measuring physical activity in Australian football. *Int J Sports Physiol Perform*. 2011;6(3):311–321. doi:10.1123/ijspp.6.3.311
18. Bates D, Mächler M, Bolker B, Walker S. Fitting linear mixed-effects models using lme4. *J Stat Soft*. 2014;67:1–48.
19. Vandenbogaerde TJ, Hopkins WG. Monitoring acute effects on athletic performance with mixed linear modeling. *Med Sci Sports Exerc*. 2010;42(7):1339–1344. PubMed ID: 20068494 doi:10.1249/MSS.0b013e3181cf7f3f
20. Montgomery PG, Hopkins WG. The effects of game and training loads on perceptual responses of muscle soreness in Australian football. *Int J Sports Physiol Perform*. 2013;8(3):312–318. doi:10.1123/ijspp.8.3.312
21. Cormack SJ, Newton RU, McGuigan MR, Doyle TLA. Reliability of measures obtained during single and repeated countermovement jumps. *Int J Sports Physiol Perform*. 2008;3(2):131–144. doi:10.1123/ijspp.3.2.131
22. Rowell AE, Aughey RJ, Clubb J, Cormack SJ. A standardized small sided game can be used to monitor neuromuscular fatigue in professional A-league football players. *Front Physiol*. 2018;9:1011. PubMed ID: 30131704 doi:10.3389/fphys.2018.01011
23. Cormack SJ, Newton RU, McGuigan MR. Neuromuscular and endocrine responses of elite players to an Australian Rules Football match. *Int J Sports Physiol Perform*. 2008;3(3):359–374. doi:10.1123/ijspp.3.3.359
24. Coad S, Gray B, Wehbe G, McLellan C. Physical demands and salivary immunoglobulin a responses of elite Australian Rules Football athletes to match play. *Int J Sports Physiol Perform*. 2015;10(5):613–617. doi:10.1123/ijspp.2014-0493
25. Jones MV, Sheffield D. The impact of game outcome on the well-being of athletes. *Int J Sport Exerc Psychol*. 2007;5(1):54–65. doi:10.1080/1612197X.2008.9671812
26. Wilson GV, Kerr JH. Affective responses to success and failure: a study of winning and losing in competitive rugby. *Pers Individ Dif*. 1999;27(1):85–99. doi:10.1016/S0191-8869(98)00226-8
27. Oliveira T, Gouveia MJ, Oliveira RF. Testosterone responsiveness to winning and losing experiences in female soccer players. *Psychoneuroendocrinology*. 2009;34(7):1056–1064. PubMed ID: 19278791 doi:10.1016/j.psyneuen.2009.02.006
28. Arruda AFS, Aoki MS, Freitas CG, et al. Influence of competition playing venue on the hormonal responses, state anxiety and perception of effort in elite basketball athletes. *Physiol Behav*. 2014;130:1–5. PubMed ID: 24642001 doi:10.1016/j.physbeh.2014.03.007
29. Young WB, Hepner J, Robbins DW. Movement demands in Australian rules football as indicators of muscle damage. *J Strength Cond Res*. 2012;26(2):492–496. PubMed ID: 22233791 doi:10.1519/JSC.0b013e318225a1c4

Authenticated t_lambrinides@yahoo.com | Downloaded 03/18/20 12:36 PM UTC